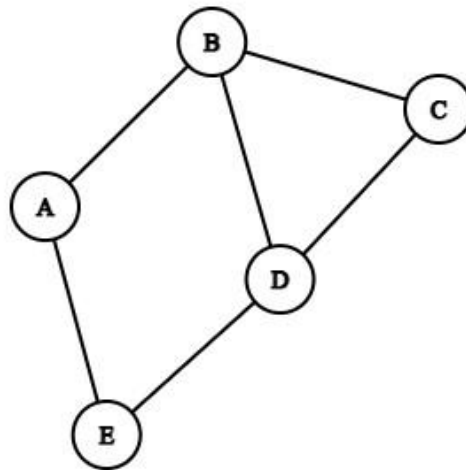


AI Lab 07 – Constraint Satisfaction Problems

Task 01: Graph Coloring

Graph coloring is a problem in which we assign colors to the vertices/nodes of a graph such that not two adjacent vertices have the same color.

Considering the following graph, implement CSP solution to find all the possible ways of graph coloring using **Red**, **Green**, and **Blue** colors.



Task 02: University Wardrobe Scheduler

Design and model a CSP to automate your dressing schedule for a five-day university week (Monday to Friday). The goal is to assign one unique outfit to each day while adhering to specific style and availability constraints.

Given Resources:

Shalwar Qamees (SQ): 2 distinct sets.

Shirts (S): 5 distinct shirts;

Pants (P): 3 distinct pants.

You must wear Shirt-Pants on Monday and Thursday.

You must wear Shalwar Qamees on Friday.

Task 03: Sudoku Game

Sudoku is a game in which we take a 2D grid and fill numbers in each cell in a way that not any row, column, or sub grid contains duplicate numbers as show below:

		6	2		5
			4	6	
	1	2			
5	6				4
		4	3		2
3			5		6

1	4	6	2	3	5
2	3	5	4	6	1
4	1	2	6	5	3
5	6	3	1	2	4
6	5	4	3	1	2
3	2	1	5	4	6

Model the solution using CSP.